

Wenxin ZHANG

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RESEARCH INTERESTS	My research focuses on dynamic resource allocation, especially for large-scale computing systems that power modern machine learning. I build application-grounded models and design simple, deployable algorithms with provable performance guarantees, drawing tools from stochastic modeling, optimization, and control theory.	
EDUCATION	Columbia University, Graduate School of Business • Ph.D. candidate in Decision, Risk, and Operations • Advisors: Santiago R. Balseiro, Will Ma	New York, NY 2021–2026 (expected)
	Tsinghua University • B.E. in Industrial Engineering • Minor in Data Science and Technology	Beijing, China 2017–2021
JOURNAL PUBLICATIONS	J3. “Feature-Based Dynamic Matching,” with Yilun Chen, Yash Kanoria, and Akshit Kumar. <i>Operations Research</i> , forthcoming. [Link] • Finalist, Michael H. Rothkopf Junior Researcher Paper Prize Competition , 2025 (Entrant: A. Kumar, Winner TBD) • First Place, Jeff McGill RMP Best Student Paper Prize , 2024 (Entrant: A. Kumar) J2. “Dynamic Pricing for Reusable Resources: The Power of Two Prices,” with Santiago R. Balseiro and Will Ma. <i>Operations Research</i> , 2025. [Link] J1. “A Gilmore-Gomory-Type Construction of Integer Programming Value Functions,” with Seth Brown, Temitayo Ajayi, and Andrew J. Schaefer. <i>Operations Research Letters</i> , 2021. [Link]	
REFEREED CONFERENCE PUBLICATIONS	C3. “Tail-Optimized Caching for LLM Inference,” with Yueying Li, Ciamac Moallemi, and Tianyi Peng. Neural Information Processing Systems (NeurIPS) , 2025. C2. “Distributed Load Balancing with Workload-Dependent Service Rates,” with Santiago R. Balseiro, Robert Kleinberg, Vahab Mirrokni, Balasubramanian Sivan, and Bartek Wydrowski. 26th ACM Conference on Economics and Computation (EC) , 2025. [Link] C1. “Feature-Based Dynamic Matching,” with Yilun Chen, Yash Kanoria, and Akshit Kumar. 24th ACM Conference on Economics and Computation (EC) , 2023. [Link]	
WORKING PAPERS	W4. “Distributed Load Balancing with Adversarial Arrivals” with Santiago R. Balseiro, Robert Kleinberg, and Balasubramanian Sivan. W3. “Distributed Load Balancing with Workload-Dependent Service Rates” with Santiago R. Balseiro, Robert Kleinberg, Vahab Mirrokni, Balasubramanian Sivan, and Bartek Wydrowski. [Link]	

- Under Review, *Operations Research*.
- **Finalist, Applied Probability Society Best Student Paper Competition, 2025** (Winner TBD)

W2. “Tail-Optimized Caching for LLM Inference” with Yueying Li, Ciamac Moallemi, and Tianyi Peng.

- Accepted at 3rd ICML Workshop on Efficient Systems for Foundation Models, 2025.

W1. “Leveraging Offline Data for Online Decision-Making in Bayesian Multi-Armed Bandits,” with Santiago R. Balseiro and Will Ma.

EXPERIENCES	Student Researcher , Google Research NYC	Summer 2024 & 2025
	• Hosted by Balasubramanian Sivan. Worked on distributed load balancing.	
	Data Science Intern , ParkHub	Fall 2023
	• Designed algorithms for real-time pricing using historical data and booking trends.	
	UGVR Research Assistant , Stanford University	Summer 2020
	Visiting Undergraduate Researcher , Rice University	Summer 2019
HONORS AND AWARDS	Paul and Sandra Montrone Doctoral Fellowship	2025–2026
	Finalist, Applied Probability Society Best Student Paper Competition	2025
	Deming Doctoral Fellowship	2025–2026
	Doctoral Fellowship, Columbia Business School	2021–2026
	Tsinghua University Outstanding Undergraduate Student Award	2021
	Beijing Outstanding Undergraduate Student Award	2021
	National Scholarship, Ministry of Education of China	2020
	Baosteel Scholarship, Baosteel Education Fund	2020
TEACHING EXPERIENCE	Columbia University, Teaching Assistant	
	• Operations Management (EMBA Core)	Fall, Spring 2024
	• Revenue and Supply Chain Management (MBA)	Summer 2024
	• Managerial Statistics (EMBA Core)	Fall 2023, 2022
	• Managerial Statistics (MBA Core)	Fall 2023
	• Healthcare Management, Design, and Strategy (MBA)	Fall 2022, 2021
SELECTED TALKS	<i>Distributed Load Balancing with Workload-Dependent Service Rates</i>	
	• The Chinese University of Hong Kong, Shenzhen, School of Data Science, Shenzhen	September 2025
	• Next-Gen Scholar’s Symposium, Singapore	September 2025
	• Third Purdue Operations Conference, West Lafayette	August 2025
	• Economics and Computation (EC) conference, Stanford	July 2025
	• INFORMS Applied Probability Society Conference (APS), Atlanta	July 2025
	• Marketplace Innovation Workshop (MIW), virtual	May 2025
	• NYC Operations Day PhD Colloquium, New York	April 2025
	<i>Leveraging Offline Data for Online Decision-Making in Bayesian Multi-Armed Bandits</i>	
	• INFORMS Annual Meeting, Seattle	October 2024
	<i>Dynamic Pricing for Reusable Resources: The Power of Two Prices</i>	
	• POMS-HK International Conference, Hong Kong	Jan 2025

- INFORMS Annual Meeting, Phoenix Oct 2023
- Revenue Management and Pricing (RMP) conference, London July 2023
- Manufacturing & Service Operations Management Conference (MSOM), Montreal Jun 2023
- Marketplace Innovation Workshop (MIW), virtual May 2023
- NYC Operations Day PhD Colloquium May 2023

PROFESSIONAL SERVICES	Co-organizer for the NYC Operations Day PhD Colloquium Student social co-chair for the Decision, Risk, and Operations division	2023 2022-2024
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REFERENCES	Prof. Santiago R. Balseiro George E. Warren Professor of Business Decision, Risk, and Operations Columbia Business School E-mail: srb2155@gsb.columbia.edu Phone: +1 (646) 460-1264 Prof. Tianyi Peng Assistant Professor Decision, Risk, and Operations Columbia Business School E-mail: tianyi.peng@columbia.edu Phone: +1 (617) 230-1062 Dr. Balu Sivan Research Scientist Google New York E-mail: balusivan@google.com Phone: +1 (212) 565-7760	Prof. Will Ma Roderick H. Cushman Associate Professor Decision, Risk, and Operations Columbia Business School E-mail: wm2428@gsb.columbia.edu Phone: +1 (617) 938-9151 Prof. Bobby Kleinberg Professor of Computer Science Cornell University E-mail: rdk@cs.cornell.edu Phone: +1 (607) 255-9200
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